Independent study — report through Week 9

physio-placebo

Is the model reading the recording, or the words around it?

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Covers	the work through Week 9
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Public plan	`https://osf.io/62r5t` DOI 10.17605/OSF.IO/62R5T

Recent papers report that a language model can take a wearable recording and judge stress or mental effort. The recording is rarely shown on its own. It is first turned into sentences: heart rate is high, the skin looks sweaty, the pattern is the stress pattern. Those sentences already suggest the label. The open question is whether the model is using the recording.

The project name refers to that possibility. A placebo resembles the treatment. Here the prompt can resemble evidence.

The experiment uses public lab recordings in which the label is the assigned task. Each recording is cut into minutes. Each minute is shown to a language model with public weights, in three forms: the numbers, short sentences about heart and skin, and a line plot. The recording is then replaced by noise, by another person, or by nothing. A score that remains high does not depend on that minute of physiology.

On the stress recordings, a simple logistic regression on ordinary heart and skin features scores 0.92. The best language model, on the real minute, scores 0.60. A plot of the trace does no better than guessing, which is about 0.41 here, because always picking the common class is not a coin flip when the classes are uneven. Noise does not always land on that line. On the averaged heart trace, Qwen's noise score is 0.48, which is what a model that answers both letters about equally often gets from this score. A flat line and a blank prompt do sit on guessing, and so does another person who was doing something else. A scrambled trace that keeps the spectrum still carries the answer for the linear model, because several of the features are the level of the trace and the size of its wiggles, which a phase scramble does not remove. A recording from a different person who was under the same kind of stress carries it too. When the model is reading sentences, it follows whatever class information those sentences still contain. That sentence test has now been run. On WESAD, Qwen loses little when the judgment words are removed, and loses a great deal when those words are flipped. The numbers are in the sentence section below.

A second experiment was requested at the last meeting: a comparison with a recent paper on matched terms. Same public data, same kind of model, prompting beside prompting, and fine-tuning beside fine-tuning. That paper uses daily self-ratings of stress from 1 to 5, together with step counts and sleep. It does not use these lab minutes. The comparison is reported separately, in Section 9, so the two scores are not placed in one table.

The comparison from the last meeting is Section 9. Same published dataset, same kind of model. Prompting in one table, fine-tuning in another. It does not use WESAD, MAUS, or CogWear.

1 Scoring

Each minute is one question. The model does not write a paragraph. I look at two letters and take the one it considers more likely. `A` is the calmer class. `B` is the harder one. On WESAD that is calm versus the speech. On MAUS it is the easy memory round versus the hard one. On CogWear it is sitting quietly versus the color-word test.

The wording of the question was chosen on two people per dataset and then left alone. Those two people are not in the scores. When a sentence says a value is high or low, that word is set from everyone else, never from the person being scored. The language models are not trained on these recordings. The only trained model in the report is in the brown section, because that paper trained one, and the request was to compare a fine-tune with a fine-tune.

The same procedure is applied to each dataset. A range in brackets is a 95% interval from redrawing the people one thousand times. The resampling unit is the person, so one long recording does not dominate the interval.

2 The score

Every wearable score in the tables is macro-F1: the average of the two per-class F1 scores, not the average of the two hit rates. The hit-rate average is balanced accuracy, and it is exactly 0.5 for any answer that ignores the label. Macro-F1 is not. On WESAD most minutes are calm, so always saying "calm" scores about 0.41, and always saying "stress" scores about 0.23. A model that says each letter about half the time lands near 0.48. I still call 0.41 guessing when I mean the common-class answer. It is not 0.50, because the two classes are not the same size, and because this score moves when the model changes how often it says the rare class.

The logistic regression is the one place a model is fit on these recordings. It is trained on everyone except the person being scored, tested on that person, and this is repeated until each person has been left out once. That is the fair way to ask whether the features carry the class for a new person, rather than for a person the model has already seen.

3 The recordings, and why these ones

I wanted a label I could point to in the lab protocol. Stress means the person was doing the stressful task. It does not mean they wrote "I feel stressed" in a diary. If that sentence is already in the prompt, the heart trace is optional, and the check is over before it starts.

The files also had to be public. Anyone should be able to download the same recordings and rerun this. A private watch corpus cannot be checked, and I cannot see whether a result used the pulse or some other field that was already the answer.

Three tasks, not one. The claim in the papers is not "this one dataset." It is stress, and mental workload, and mental effort. If I only ran the dataset where everything is easy, a success would be a success on that dataset. WESAD is the one the field already uses for lab stress, and a simple model solves it, so a language model near guessing there is failing a test that can be passed. MAUS and CogWear are in the study so I cannot sell a WESAD result as a general one. On those two the simple model is weak. That is useful. It means there is less in the recording for anyone to read, and a language-model score near guessing is not a surprise.

This is what one recording looks like. A person sits in a lab, wearing the sensors, doing a timed task. I take 60 seconds. The next piece starts 30 seconds later, so neighboring minutes share half their samples. A piece never crosses from the task into the rest period. The next section

prints one of those minutes the way the model actually receives it. A few seconds of the raw heart trace show up again when I start swapping the recording out.

WESAD is fifteen people from the public stress dataset released by the University of Siegen. They wear a chest belt and a wristband. The belt records the electrical heart trace, skin conductance (how easily the skin passes a tiny current, which rises with sweat), and breathing, 700 times a second. The wristband records pulse with a light, 64 times a second, and skin conductance more slowly. The hard class is a public speaking task, the Trier Social Stress Test. The calm class is sitting quietly, plus watching a funny clip. The files also contain a meditation block. I leave it out. It is not the speech, and it is not the quiet sitting or the funny clip. Motion, muscle activity, and temperature are in the files too. I do not use them. The question is heart, sweat, and breath.

MAUS is twenty-two people doing a memory game, the n-back, with a wrist pulse sensor. In the easy rounds (0-back) they respond to the item on the screen now. In the hard rounds (2-back and 3-back) they have to remember what came two or three items ago. The label is that difficulty, not a stress rating. The files also contain a chest heart trace and a fingertip pulse. I use the wrist pulse. That is the wearable in the claim, and it keeps this dataset from becoming a second chest-ECG study. Resting recordings are in the release and are not part of the task, so they are out.

CogWear is a wristband study from PhysioNet. The hard class is the Stroop test: the word "blue" printed in red ink, and the person has to name the ink. The other class is sitting quietly. I use the pilot group, ten people with both pulse and skin conductance. An eleventh person is missing the skin-conductance file in the public release, so that person is out. A second cohort in the same archive played a different protocol, so I did not pool them together. A headband EEG is in the files. I leave it out. A headset is a different question from a watch.

I did not add daily self-ratings as a fourth wearable task. People scoring their own day from 1 to 5 is the brown section, on a dataset called PMData, because a published paper did that experiment and I was asked to meet it on its own terms. Private industrial corpora, including the Fitbit data behind SensorLM, are not here. There is nothing public to download.

Dataset	People included	Minutes	What the two classes are
WESAD	15 of 15	1,042	731 calm / 311 stress
MAUS	22 of 22	1,086	362 easy / 724 hard
CogWear	10 of 11	114	39 rest / 75 Stroop

No minute was dropped for a missing sample. From each minute I compute the same kind of summary a standard physiology toolbox would: heart rate, how much the time between beats varies, skin conductance, and breathing where the belt recorded it. The toolbox is NeuroKit2, version 0.2.13, and I have not edited it. The logistic regression and the sentences are built from those same summaries.

4 What a model is shown

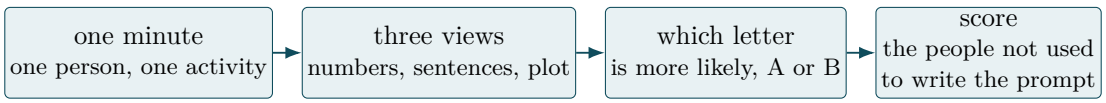


Figure 1: What happens to one minute. The wording is chosen on two people and then left alone. Every score below is the other people.

Here is one real WESAD minute, in the three forms. The numbers are the chest heart trace, called an ECG in the prompt, averaged into 128 values so a prompt can hold it. The sentences are four of the eighteen lines the model actually sees. “eda” is skin conductance. “hrv” is heart-rate variation. Words like “stress-typical” are deliberate. The sentence test later in this report takes those words out, and then reverses them. The plot is a drawing of that same minute, with no name and no label on it.

Numbers — the ECG averaged into 128 bins

```
ecg fs=700 Hz n=128 scheme=bin_mean:
0.04024 0.005173 -0.01352 -0.01912 0.03245 -0.00422 -0.02569 0.0196
... 117 values omitted ...
-0.04705 -0.00251 -0.06615
```

Sentences — 4 of the 18 feature lines

```
eda_tonic_mean is reduced (0.7358), calmer-typical.
hrv_mean_hr is elevated (89.28), stress-typical.
hrv_sdnn is reduced (41.18), stress-typical.
hrv_rmssd is reduced (17.7), stress-typical.
```

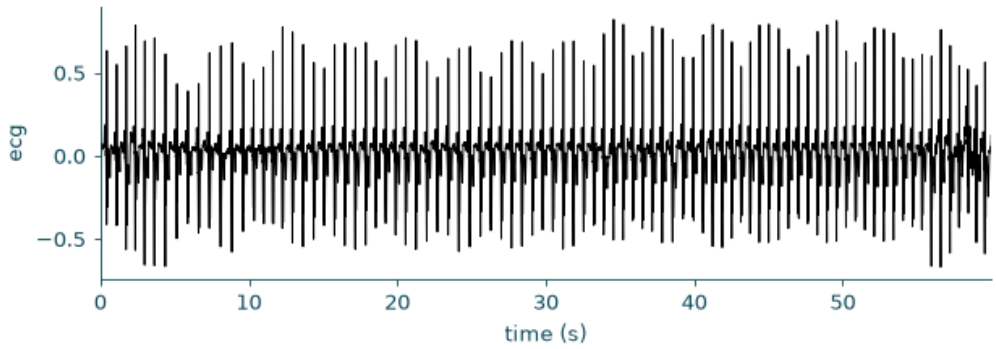


Figure 2: The drawing the vision model receives for the minute quoted above. Nothing on the figure says who it is, or which class it is.

5 A simple model, before any language model

I wanted a baseline that does not involve a language model at all. Before any of those runs, I fit a logistic regression and a small decision tree on the features. If a straight line through heart rate and skin conductance can separate the classes, then the information is in the recording. A language model that scores near guessing is leaving that information unused.

The shuffle ceiling is a check against luck. I shuffle the labels inside each person, refit a thousand times, and take the score that 95% of those shuffles fall below. A real result has to clear that line.

Dataset	Logistic regression	Small tree	Guessing	Shuffle ceiling	Useful?
WESAD (15 people)	0.918	0.848	0.412	0.433	Yes
MAUS (22 people)	0.444	0.482	0.400	0.413	Barely
CogWear (10 people)	0.569	0.560	0.397	0.541	Barely

WESAD is the dataset that can answer the question. The linear model is far above guessing, so there is something for a language model to miss. MAUS and CogWear are weak even for

the linear model. On those two, a language-model score near 0.4 does not mean much. When I later compress each result into one ratio, those two will often be left blank, and I will show the raw gap instead.

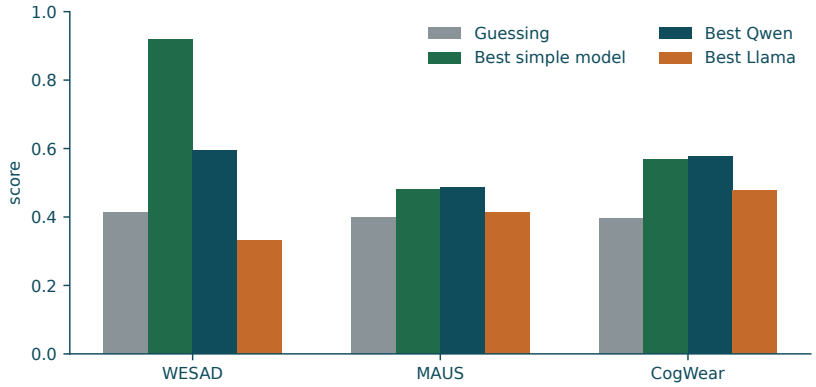


Figure 3: The best simple model against the best language-model score on the real recording. Each language-model bar is that model's best of the three ways of showing a minute. WESAD is the only dataset where the simple model is far above guessing. The score is the average of the two class hit rates, explained above.

6 Language models on the real recording

The text models are Qwen3-8B and Llama-3.1-8B. Both are open weights, which means anyone can download them. The public plan originally named GPT-4.1. I switched, because that account was capped at 50 requests a day. The plan has a public note saying so.

"Every nth sample" keeps one sample out of every few, so the minute fits in a prompt. "128 averages" splits the minute into 128 bins and averages inside each bin. "Feature sentences" are the lines from the previous section. The two people set aside for wording are WESAD S2 and S8, MAUS 015 and 023, and CogWear 4 and 8. Every minute in this table produced a score.

Dataset	What the model sees	Qwen3-8B	Llama-3.1-8B
WESAD	every nth sample	0.427 [0.392, 0.465]	0.333 [0.301, 0.370]
WESAD	128 averages	**0.595 [0.550, 0.647]**	0.230 [0.227, 0.233]
WESAD	feature sentences	0.584 [0.462, 0.699]	0.283 [0.247, 0.324]
MAUS	every nth sample	0.467 [0.436, 0.499]	0.413 [0.403, 0.426]
MAUS	128 averages	0.483 [0.452, 0.516]	0.400 [0.400, 0.400]
MAUS	feature sentences	0.487 [0.439, 0.531]	0.400 [0.400, 0.400]
CogWear	every nth sample	0.576 [0.443, 0.689]	0.404 [0.392, 0.417]
CogWear	128 averages	0.396 [0.380, 0.411]	0.440 [0.399, 0.495]
CogWear	feature sentences	0.475 [0.393, 0.504]	0.479 [0.375, 0.640]

The bold row is the best language-model result so far. Qwen, on WESAD, reading the heart trace as 128 averages, with no examples in the prompt, scores 0.595. The range sits above guessing. It does not come close to the logistic regression at 0.918.

The feature sentences are close behind, at 0.584 [0.462, 0.699]. The range starts above guessing and is wide. The sentence section scores this same prompt again. That pass is 0.581 [0.460, 0.695] for Qwen and 0.278 [0.245, 0.316] for Llama, against 0.283 [0.247, 0.324] here. The two passes agree in the second decimal.

Llama's two MAUS rows sit at 0.400 with a range that has no width. That is not a crashed

run. On every person in the table, Llama gave the same answer every time. CogWear's ranges are wide because only 90 minutes are scored.

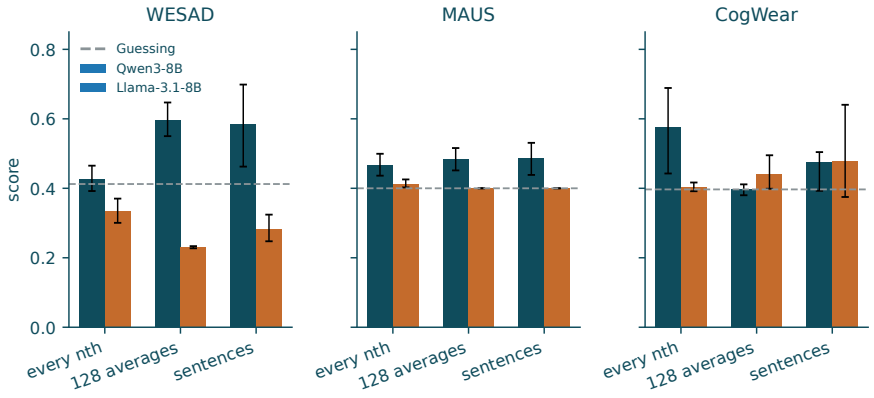


Figure 4: Qwen in dark teal, Llama in orange, on the real recording. The dashed line is guessing. The bar that clears it, with the whole range above the line, is Qwen on WESAD with the averaged heart trace.

7 A drawing, instead of numbers

The third model, Qwen2.5-VL-7B, sees a line drawing. On WESAD that is the chest heart trace. On MAUS it is the wrist pulse. The drawing has no name and no class written on it. The same minute always makes the same file. CogWear has no drawing condition. The prompt includes four example drawings, chosen before these scores were computed.

Dataset	What was in the prompt	Score	Minutes
WESAD	four example drawings	0.412 [0.410, 0.414]	906
MAUS	four example drawings	0.400 [0.399, 0.400]	990

Both sit on guessing. The ranges are tight because the model almost always picks the same class. The prompt was not revised after these scores.

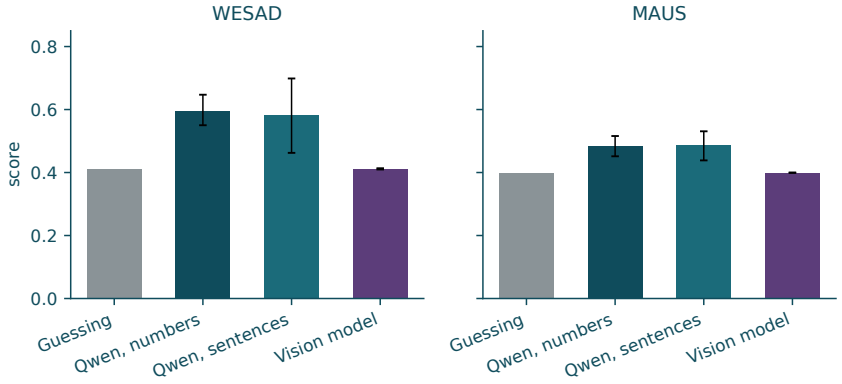


Figure 5: Guessing, Qwen's best number version, Qwen's sentences, and the vision model. The drawing does not beat guessing on WESAD or on MAUS.

8 Replacing the recording

The prompts stay as they were. For each minute I swap the trace and score again. Five swaps are there to remove the class. One is there for a different reason.

- **Gaussian noise.** Random noise with the same average and spread as that sensor across the whole dataset. Stress and calm are not given their own noise.
- **Phase-randomized.** The sizes of the wiggles stay the same. That is the spectrum. Their timing is scrambled, so it no longer looks like that person's heartbeat.
- **Wrong person, wrong activity.** A minute from someone else, and I do not force it to be the same activity.
- **Flat line.** Every sample is replaced by the average of that sensor.
- **Nothing left.** The prompt says the signal was removed, and there are no digits left in it.
- **Same activity, other person.** Another person who was doing the same thing. This one is not trying to destroy the signal. It asks whether the model needs this recording, or whether stressed people simply look like each other.

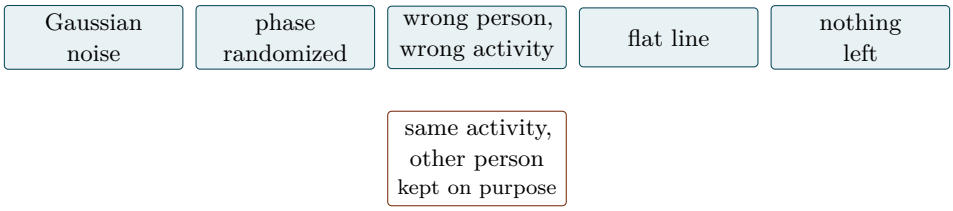


Figure 6: Five ways of removing the class, and one swap that keeps it. The lower box is not a destroyed signal. It is a different question.

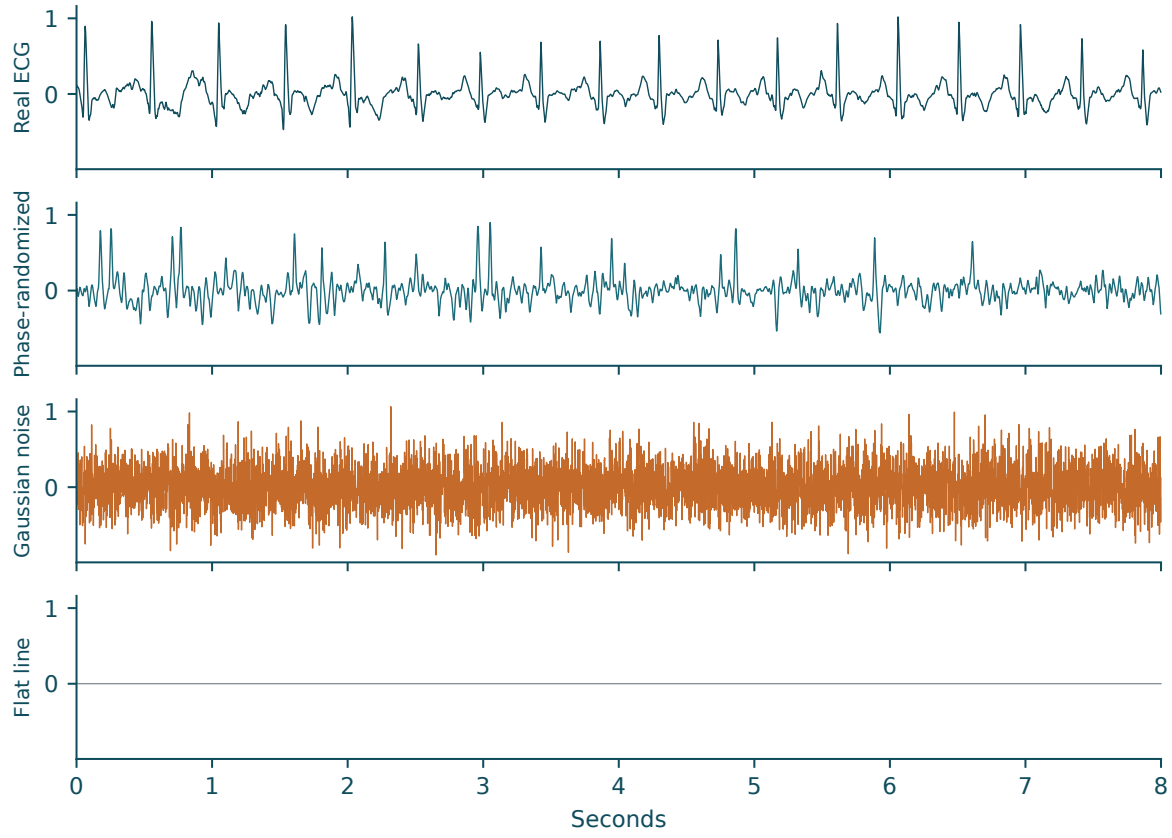


Figure 7: Eight seconds of chest ECG from one stress minute, WESAD person S10. This is a different minute from the prompt example above, chosen so the four traces can share a scale. Top is the recording. Below it: phase-randomized, Gaussian noise, and a flat line at the same average. The person and the label stay. Only the trace changes. On a shared scale the flat line is actually flat.

I scored every setup from the real-recording experiment on every replacement. That is 120 runs, and none of them failed. The probability of each letter is saved with every minute.

Before looking at the language models, I put the same features through the same logistic regression. This says how much class information each replacement still holds. I measure that. I do not assume a replacement is empty.

Replacement	WESAD	MAUS	CogWear
Gaussian noise	0.412	0.400	0.486
Phase-randomized	0.880	0.405	0.458
Wrong person, wrong activity	0.411	0.405	0.380
Flat line	0.412	—	0.397
Signal deleted	—	—	—
Same activity, other person	0.954	0.494	0.773

A dash means there was nothing to fit. A flat pulse on MAUS has no heartbeats, so the features come out empty. Deleting the signal leaves no features on any dataset.

The WESAD column is the one that says something. Noise, a flat line, and the wrong person in the wrong activity all fall to 0.41, which is the common-class score. A phase-randomized trace is still at 0.88. Another person doing the same thing is at 0.95. That is not because the class lives in the spectrum. Phase randomization keeps the value distribution of each minute, so the level of skin conductance and the size of the wiggles stay put. Those are the features the linear model is using. The correlation between a real WESAD minute and its phase-randomized copy is 1.00 for the skin-conductance level and 0.77 for mean heart rate. MAUS and CogWear never had a strong linear score, and swapping the trace does not create one.

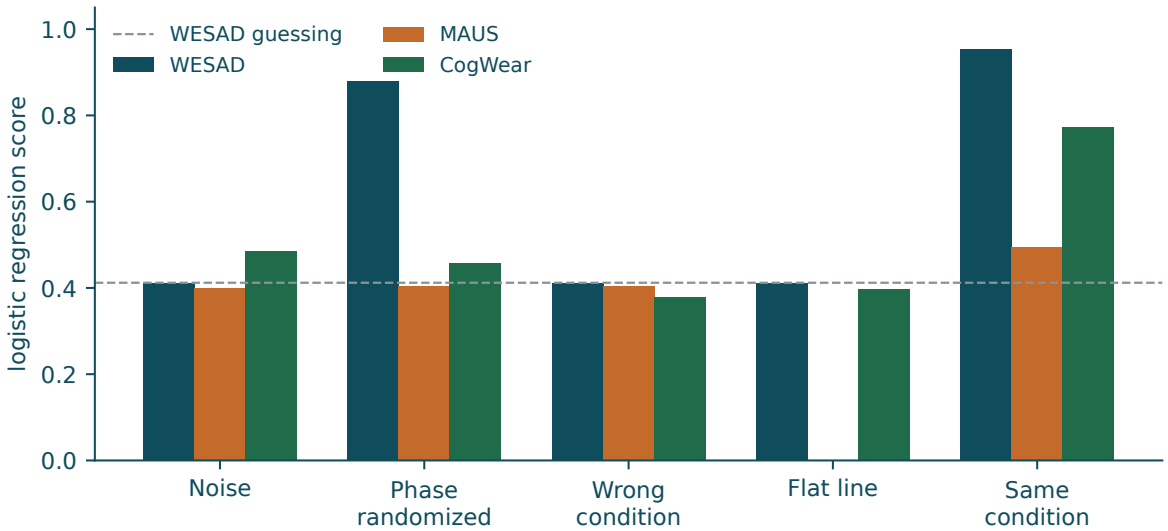


Figure 8: Logistic regression on each replacement. The missing orange bar, MAUS with a flat line, is the empty-feature case in the table, not a score of zero. The dashed line is guessing on WESAD.

The same replacements then went to the language models. The chart is Qwen on WESAD, which is the only place a language model had clearly cleared the common-class line on the real recording. I did run Llama on the replacements as well. Its macro-F1 on the real WESAD sentences is 0.28, under the common-class line, because it answers “stress” on almost every minute. The ordering in its probabilities is not empty: the margin between the two letters has an AUROC of 0.82 on those same minutes. Macro-F1 does not show that.

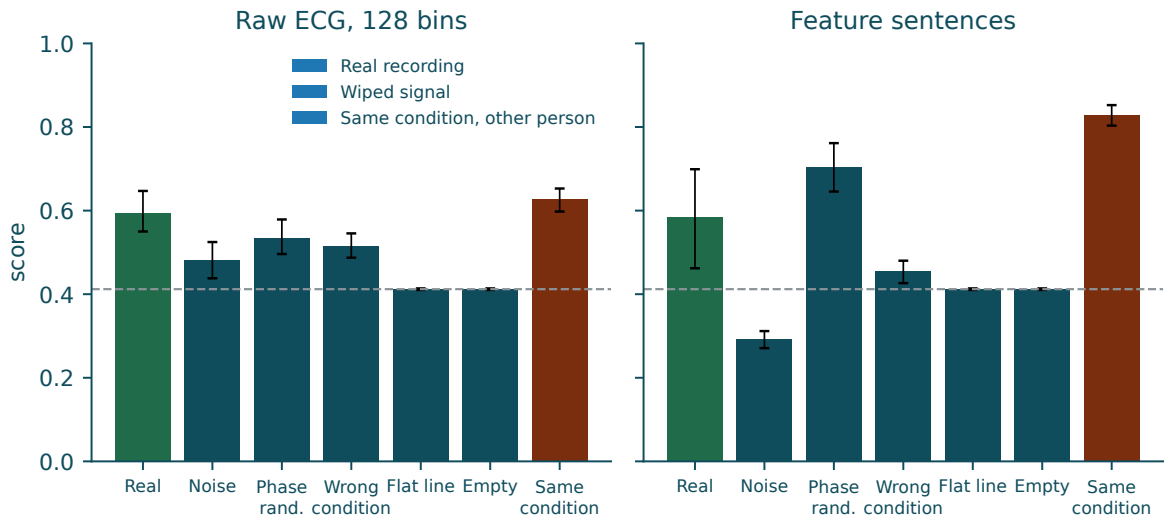


Figure 9: Qwen3-8B on WESAD, people held out of the prompt wording. Left: the heart trace as 128 averages (0.60 on the real recording). Right: the feature sentences (0.58 on the real recording). Green is the real recording. Teal is a replacement meant to remove the class. Brown keeps the activity and changes the person. The legend is on the figure.

The right-hand panel is the sentence condition. On the real recording Qwen scores 0.58. Sentences built from a phase-randomized trace score 0.70, above the real recording. That rise is a shift in how often Qwen says "stress", not a better reading of the minute. The ordering of the two letters is weaker on the scrambled trace (AUROC 0.87) than on the real one (AUROC 0.94). Sentences from another stressed person score 0.83, and the ordering stays high (AUROC 0.92). Sentences built from noise fall to about 0.29, below the common-class line, and the ordering goes to chance. A flat line and an empty prompt sit on the common-class score. The left-hand chart, the raw trace, is the one where noise does not: Qwen's noise score there is 0.48.

On the sentences, this agrees with the linear model in one respect. When the sentences still contain the class, Qwen scores well even if the waveform is no longer that person's. When the sentences are noise, or gone, Qwen falls to the common-class line or below it. The raw trace is a smaller effect, and noise there does not fall to 0.41.

9 What you asked for

The answer, before the tables.

The paper that matches the request is Health-LLM, from CHIL 2024. Their task is a daily stress rating from 1 to 5. Lower error is better.

Prompting their model: they report 0.76. On the raw day-by-day list, this rerun is 2.69, because the model says "out of 5" and the scoring rule writes down 5. On their short prompt, the 14-day average, with the sampling decoder their appendix calls the default, the error is 0.84, inside the band around 0.76.

Qwen, on their prompt, is at 0.80. Close to their prompting result, and worse than always answering 3.

Fine-tuning every weight of their model on their data, with their 7B recipe, brings the error to 0.44. Their paper says 0.38. An earlier run that trained only a small add-on barely moved, and is kept as a record.

None of these numbers are wearable scores.

At the last meeting you asked me to find recent papers, from a real venue, and to compare like

with like. If a paper only prompted the model, that result sits next to prompting. If a paper fine-tuned, that result sits next to a fine-tune of the same kind of model, on the same public data. A classification score on WESAD does not belong next to an error on a 1-to-5 stress rating. This section is only the second kind of number.

SensorLM, from NeurIPS 2025, is about wearables and language models, but the weights and the Fitbit recordings are not public, so it is not in the tables. Other recent papers keep their labels private, or they are not language models. I left those out for the same reason. Every paper in that reading, and the earlier papers this study is written against, is named in Appendix A.

9.1 Their task

PMData is a public daily-life dataset. People wore a consumer tracker and, each day, rated their stress from 1 to 5. The model sees two weeks of steps, calories, resting heart rate, sleep, and that day's mood, and it has to give that day's rating. The error is the average distance from the person's own number. Always answering 3, on the 299 days I use as the test, is off by 0.401. That constant prediction is the reference for the errors below.

The model they released is MedAlpaca-7B. Their Table 3 is prompting, with no examples in the prompt. Their Table 4 is the same model after it has been fine-tuned on PMData. I kept those as two tables.

9.2 Prompting only

I used their prompt, and a test set of 299 days that was fixed before the run. The scoring rule was fixed too: read the first number in the answer. I did not change the rule when the number looked wrong.

A lot of MedAlpaca's answers say "out of 5" or "out of 10." The rule records 5 or 10. A stress rating only goes to 5, so those answers are far from a typical day. That is why this rerun sits at 2.69 while the paper says 0.76. I left the rule as it was.

Their own paper also writes the two weeks as one average per sensor, not as the raw list, and the appendix sets sampling as the default decoder. I ran that prompt, temperature 0.4, three seeds, same first-number rule, same 299 days. The errors are 0.823, 0.853, and 0.846. The mean is 0.841, and the spread is 0.016. Their published band is 0.76 ± 0.1, so 0.841 is inside it. It is not the same point as 0.76. The model says 2 or 3, not "out of 5."

Qwen, given the raw list, answers "Answer: 3." Its error is 0.795, near the paper's prompting result, and still worse than the constant answer of 3.

Llama, given the same words, does not answer with a stress rating. It writes a short Python program, and the rule reads a step count out of that program. The error is about 2,650. The scoring rule was not changed.

Row	Model	What was done	Average error
Their Table 3	MedAlpaca-7B	no examples	0.76 ± 0.1
This rerun, raw list	MedAlpaca-7B	day-by-day list, greedy	2.689 ± 0.038
This rerun, their average	MedAlpaca-7B	14-day average, sampling	0.841 ± 0.016
Our model, their prompt	Qwen3-8B	no examples, same rule	0.795 ± 0.020
Our model, their prompt	Llama-3.1-8B	no examples, same rule	2654 ± 43

Lower is better. This table is prompting. It is not a WESAD score, and it is not the fine-tuning table below.

9.3 Fine-tuning only

Their Table 4 says that fine-tuning MedAlpaca on PMData brings the error down to 0.38.

I fine-tuned the same model on the training examples they released. One training example also appeared in the locked test set, so I dropped it. The training file has 653 examples. I did not oversample, because they never published the rule for it. I also skipped their 100-step warmup. With only 653 examples at batches of 128 there are 18 update steps, and a warmup that long would never reach the learning rate.

Their 0.38 uses their own test split. My number uses the 299 days I had already locked. Same model, same task, same error. I am not claiming the two test lists are the same list.

The first attempt trained only a small adapter, an add-on of a few million weights, at their learning rate of 2×10^{-5}. It barely moved. The adapter starts at zero, and after 18 small steps its largest weight was about 0.0002. Scored on the raw question it said "out of 5" and landed at 2.697, the same as the model before training. That number stays as a record of a run that did not train.

Their appendix says the parameters themselves are updated. So the second run trains every weight: 3 epochs, batches of 128, learning rate 2×10^{-5}, cosine schedule. It is scored on the template it was trained on, which ends at "Response." The paper used four large GPUs. This machine has one 48 GB card, so the weights are stored in a 16-bit format and the optimizer keeps its running averages in 8 bits. The training loss fell from 1.84 to 1.04 over the 18 steps.

After that run, every answer reads "The predicted stress level is N." With the sampling decoder from their appendix, three seeds, the error is 0.448, 0.435, and 0.442. The mean is 0.442, and the spread is 0.007. Their paper says 0.38. Greedy decoding says 3 on all 299 days, which is exactly the constant answer, at 0.401.

Row	Model	What was done	Average error
Their Table 4	MedAlpaca-7B	fine-tuned on PMData	0.38
This rerun	MedAlpaca-7B	every weight trained, sampling	0.442 ± 0.007
This rerun	MedAlpaca-7B	every weight trained, greedy	0.401
First attempt	MedAlpaca-7B	small adapter, did not train	2.697

Lower is better. This table is fine-tuning. It does not get compared with the prompting table, and it does not get compared with WESAD.

What I take from this. On the raw list, with their model and a scoring rule fixed in advance, I do not get 0.76. The model says "out of 5." On the 14-day average, with sampling, the error is 0.84, inside their published band and not equal to 0.76. Training every weight on their data brings the error to 0.44, near their 0.38. Almost every answer after training is 3, and always answering 3 scores 0.401. Qwen, which was not trained on their data, lands near their published prompting error and still loses to always answering 3. Llama does not answer the question that was asked. This says something about running their experiment. It does not say whether a language model can read a WESAD minute.

10 The sentence test

The prompts stay the ones frozen in September. Only the sentences change. The original wording here is a second scoring of the feature-sentence row above, not a different prompt. Three wordings of the same numbers: as written, with the judgment words removed, and with those words flipped. Removing them leaves the number and its percentile among the other

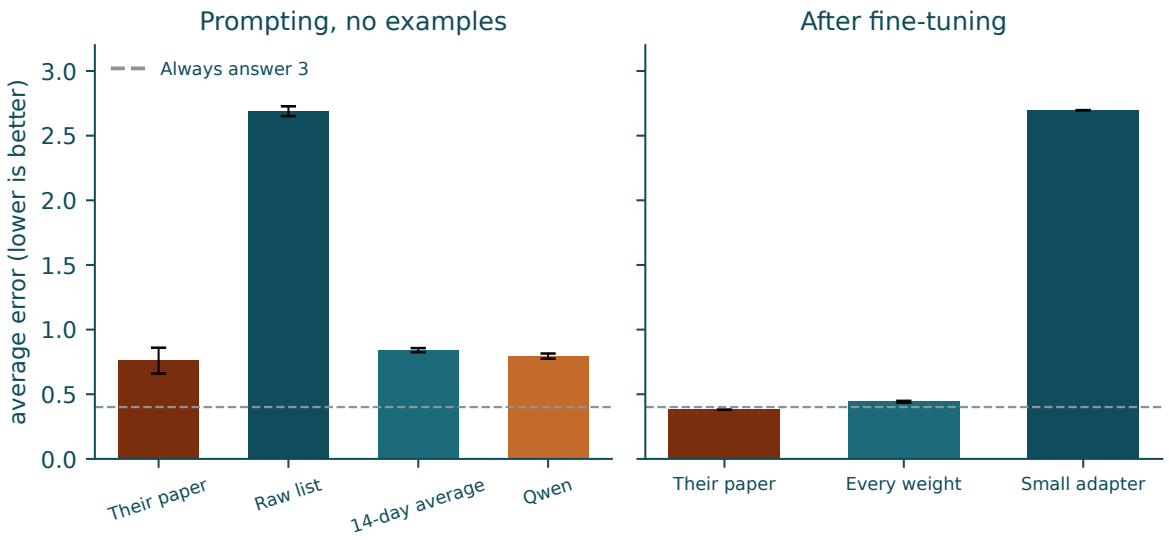


Figure 10: The two tables, drawn. Left is prompting. Right is fine-tuning. Lower is better. The dashed line is the error from always answering 3. On the left, the raw list is 2.69 and the 14-day average with sampling is 0.84. On the right, training every weight brings the error to 0.44 with sampling. Llama is not on the chart. Its error is about 2,650, because it wrote a program and the scoring rule read a step count. It is in the prompting table.

people. Flipping them swaps "elevated" with "reduced," and "stress-typical" with "calmer-typical," and leaves the number alone. Both models, all three datasets, every minute scored. No failures.

The plan filed before these runs made two predictions. Removing the judgment words should cost more than half of whatever advantage the original wording had over the shuffle floor. Flipping the words should change the predicted letter on more than 40% of minutes. If the original wording was not convincingly above that floor, the first prediction is left unscored.

Read the figure left to right. Dark is the original wording. Teal is the same numbers with the judgment words removed. Brown is those words flipped. The whiskers are the interval from redrawing people. The dashed line is guessing.

Model	Dataset	As written	Words removed	Words flipped
Qwen	WESAD	0.581 [0.460, 0.695]	0.554 [0.441, 0.662]	0.313 [0.224, 0.403]
Qwen	MAUS	0.485 [0.438, 0.529]	0.399 [0.345, 0.448]	0.507 [0.466, 0.546]
Qwen	CogWear	0.475 [0.393, 0.504]	0.542 [0.426, 0.612]	0.597 [0.513, 0.658]
Llama	WESAD	0.278 [0.245, 0.316]	0.230 [0.227, 0.233]	0.230 [0.227, 0.233]
Llama	MAUS	0.400 [0.400, 0.400]	0.400 [0.400, 0.400]	0.503 [0.450, 0.553]
Llama	CogWear	0.479 [0.375, 0.640]	0.404 [0.392, 0.417]	0.404 [0.392, 0.417]

The last column is the one to use when the question is whether flipping the words changed the letter.

Share of minutes that change	WESAD	MAUS	CogWear
Qwen	63% [52, 74]	71% [65, 77]	72% [51, 91]
Llama	3% [1, 6]	27% [15, 40]	8% [0, 18]

Qwen on WESAD loses about 17% of its margin when the words are removed. The plan said more than half, so that prediction does not hold there. On MAUS the whole margin goes against

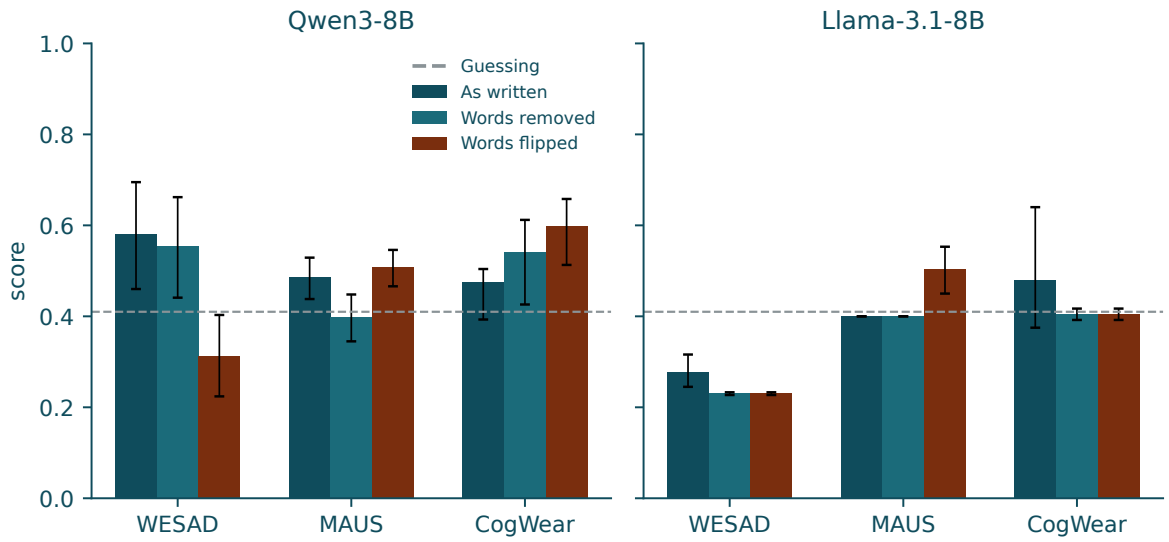


Figure 11: The three wordings. Left is Qwen. Right is Llama. Dark bars are the sentences as written. Teal bars have the judgment words removed. Brown bars have those words flipped. The dashed line is guessing.

the filed chance floor, so the filed prediction holds. A chance floor matched to how often Qwen says the hard class on MAUS is 0.499, and the original wording scores 0.485, under that floor. The MAUS result holds only under the filed floor. CogWear's original wording was not clearly above the floor, so that prediction is not scored. Flipping the words changes more than 40% of Qwen's minutes on all three datasets. It does not do that for Llama. Counting the judgment words, with no language model, already scores 0.74 on WESAD, above Qwen's 0.58, and Qwen's margin tracks that count (Spearman 0.84). SensorLM's statistical sentences were not part of this run.

11 Remaining work

The replacement charts already show what happens when the recording is swapped. The sentence test above is the second of the three filed predictions. The third, flipped words changing more than 40% of minutes, is now scored as well.

The ratio is now computed, outside this PDF, in `results/sds_card/`. In words, it is how much of the gap between the real recording and the locked chance floor disappears when the recording is replaced:

$$\frac{\text{real} - \text{replacement}}{\text{real} - \text{chance}}.$$

If the real recording was never convincingly above that floor, the ratio is left blank. Seventy-five of the one hundred null cells are blank for that reason. Where a ratio is printed, it is the filed one, including values above 1 and below 0. Those happen when the replacement scores on the wrong side of the chance floor, which this score does easily when the model changes how often it says the rare class. The item-level intervals for the raw numbers and the plots are not in the card yet. Those rows were not saved. They still have to be re-scored on the lab machine. The sentence rows were saved, and their intervals are in.

12 Where the work stands

Piece	State
Three public datasets, cut into minutes	Done. Nothing dropped for missing samples.
A simple model on the same features	Done. WESAD reaches 0.92. The other two are weak.
Language models on the real recording	Done. Best score is 0.60. Drawings sit on guessing.
Replacing the recording	Done. On WESAD the linear model collapses on noise. A phase scramble does not, because it keeps the level of the trace.
The comparison from the last meeting	Done. Section 9. Separate dataset, separate score.
Removing and flipping the judgment words	Done. Qwen on WESAD loses 17% of its margin when the words go, and 63% of minutes flip when the words reverse.
One ratio per setup	Computed. In `results/sds_card/`. Not copied into this PDF. Raw-number and plot rows still need a re-score.

A Papers I read

Two lists. The first is the reading for the comparison you asked for: recent papers on language models and wearable or daily-life health data, from 2023 through October 2026. A paper enters the brown tables only when the dataset, the task, the kind of model, and the score can sit next to Health-LLM. The second list is older. Those papers are why the wearable study exists. They are not PMData numbers, and they are not in those tables.

A.1 The comparison reading

Paper	Where it sits
Kim et al., *Health-LLM: Large Language Models for Health Prediction via Wearable Sensor Data*. CHIL 2024. arXiv:2401.06866.	In the brown tables. PMData stress, MedAlpaca-7B, prompting and fine-tuning kept apart.
Zhang et al., *SensorLM: Learning the Language of Wearable Sensors*. NeurIPS 2025. arXiv:2506.09108.	Not in the tables. Wearables and language, but the Fitbit recordings and the weights are not public.
Zheng et al., *ProMind-LLM: Proactive Mental Health Care via Causal Reasoning with Sensor Data*. arXiv:2505.14038. No conference on the arXiv record.	Out. PMData and GLOBEM, but the labels are a private psychologist coding, not the public stress rating.
Soyebo et al., *On-Device Language Models for Privacy-Preserving Stress Prediction: A Multimodal Evaluation on Mobile Health*. Preprint, 2026. arXiv:2609.11961.	Not in the tables. Same daily summaries and the same error, but a preprint, and much smaller models.
Hagos et al., *From Stress to Affect: Multimodal Deep Learning for Physiological Emotion Recognition Across Wearable Sensor Modalities*. Under review. arXiv:2609.20991.	Not in the tables. WESAD, but the models are trained classifiers, not a language model.

No conference paper in this reading fine-tunes Llama-3.1 or Qwen3 on WESAD. The wearable

scores in the body of this report are our own experiment.

A.2 Papers this study is written against

Paper	Why it is here
Jo et al., *Are EEG-to-Text Models Working?* arXiv:2405.06459.	They showed an EEG-to-text model can still look good after the brain recording is swapped for noise. Different task. Same doubt.
Tan et al., *Are Language Models Actually Useful for Time Series Forecasting?* NeurIPS 2024. arXiv:2406.16964.	They asked whether a language model actually helps at forecasting a time series. Often it does not. Not physiology. A reminder that a negative result still counts.
Xu et al., *LSM-2: Learning from Incomplete Wearable Sensor Data.* arXiv:2506.05321.	A newer Google sensor model. The paper is public. The code and the weights are not, so I cannot run it.
Zhu et al., *VitalAgent: A Tool-Augmented Agent for Reactive and Proactive Physiological Monitoring over Wearable Health Data.* arXiv:2605.29483.	A 2026 system that compares having the true heart rate against not having it. Useful, and it is not the same test as replacing the recording with noise.

WESAD, MAUS, CogWear, and PMData are the recordings, not extra papers in this list.

Shrijak Kumar • 30 October 2026 • Dr. Siddharth • Plaksha University